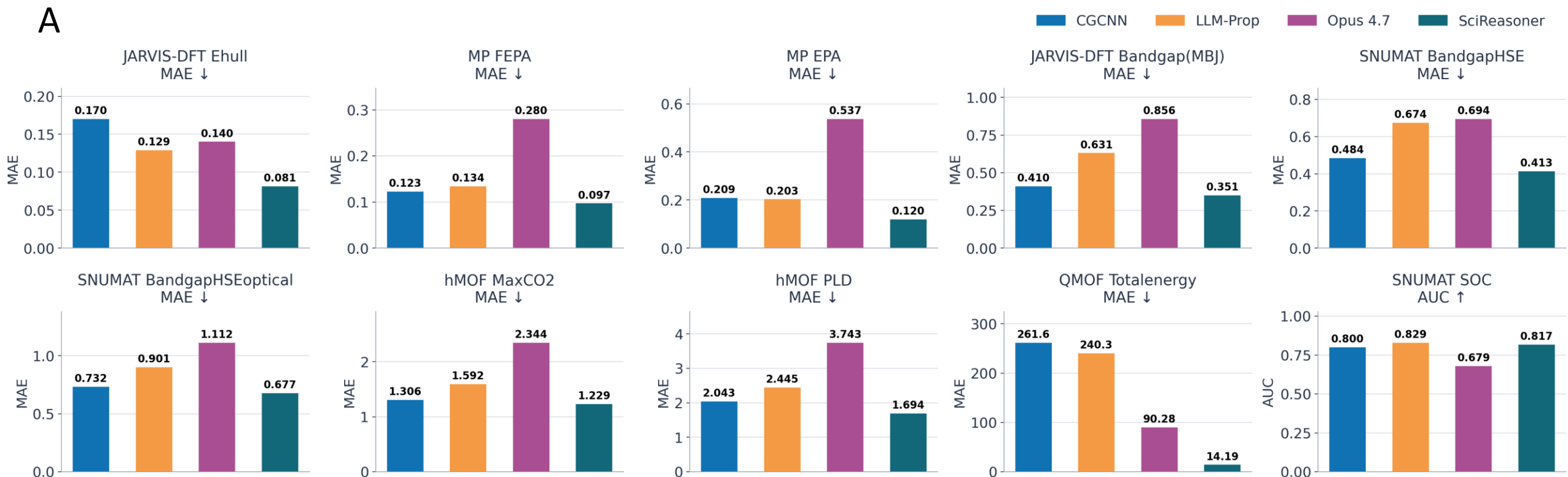
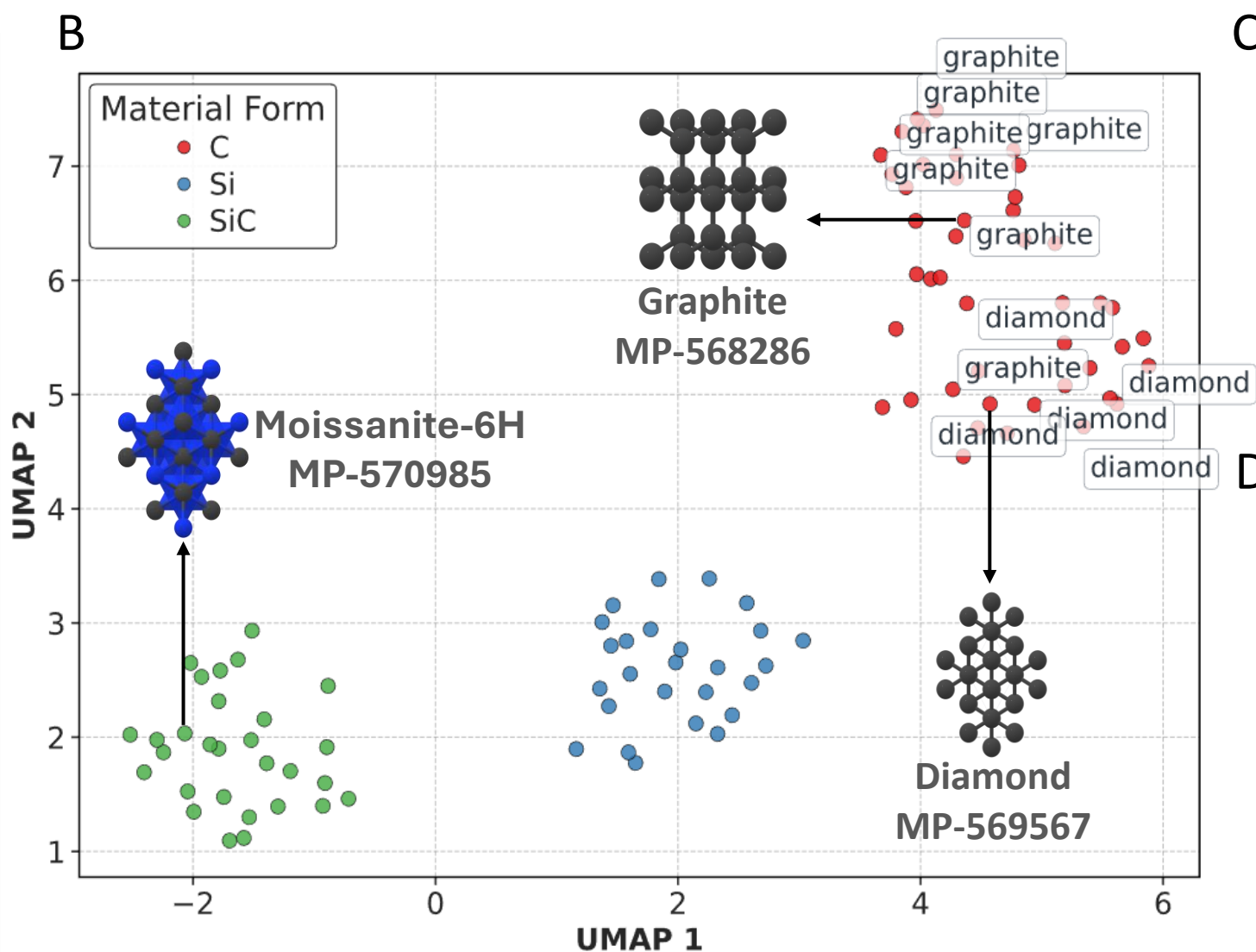


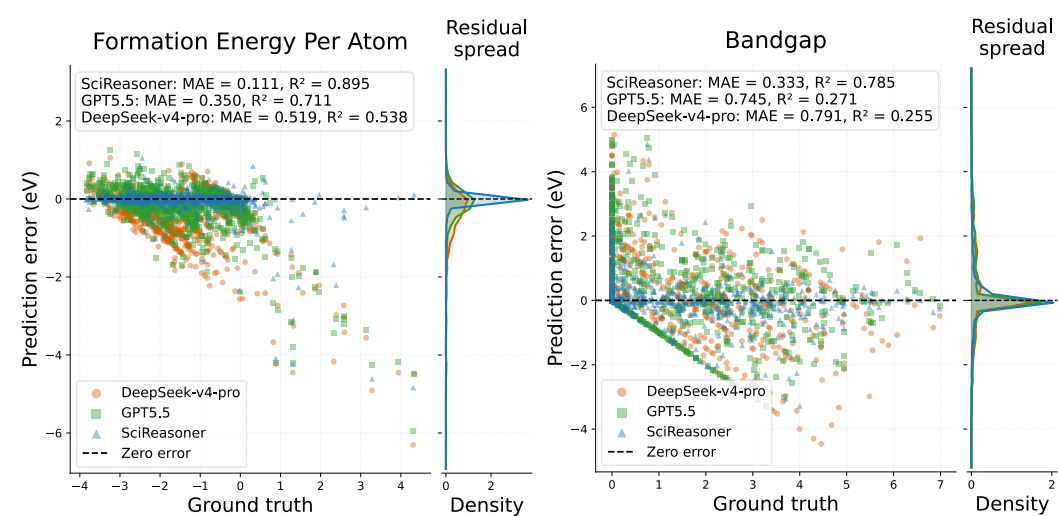
A



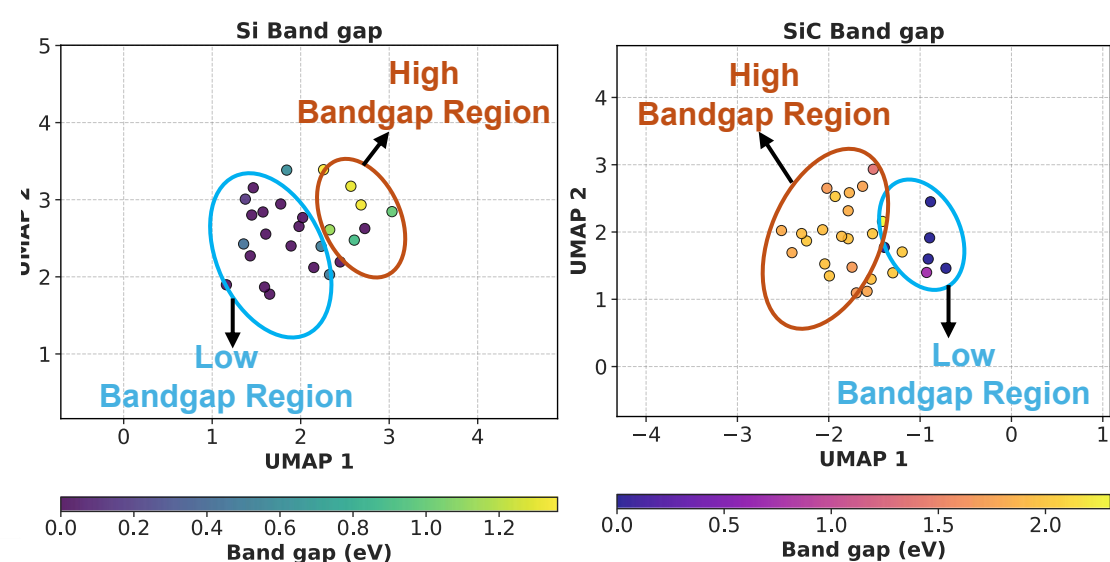
B



C



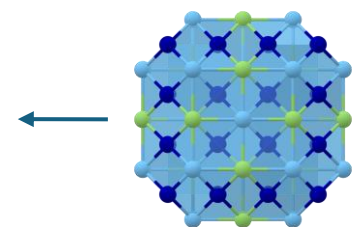
D



E



Look at the chemical formula and structure information of the given crystalline material and predict its property. Is the following material (TiGaCo2) stable?



<think>To determine the stability of the crystalline material with formula TiGaCo2, I need to analyze the provide structure ...The token sequence `<material_structure>+ t a ODD a DOD b 000 c 000 d 000 e 000 o</material_structure>` maps directly to the **space group Fm-3m, a high-symmetry system often associated with stable intermetallic compounds** ... Next, I parse the atom list to verify the stoichiometry... Then, I investigate the bonding network to understand the local coordination environment. The bond `<material_structure>0 12 --o</material_structure>` connects a titanium atom to a cobalt atom across a unit cell boundary ... `<material_structure>0 6 ooo</material_structure>` shows a titanium-gallium bond within the same cell ... The density of these edges, such as `<material_structure>1 11 o--</material_structure>` ... `<material_structure>4 9 ---</material_structure>` ... This **heteroatomic bonding topology is crucial for stabilizing the structure through electronic interactions** ... The combination of the Fm-3m space group, the 1:1:2 stoichiometry, and the fully connected bonding network is characteristic of the L2₁ Heusler structure type. Therefore, the structural evidence points strongly towards a stable material configuration. </think>

+ t a ODD a DOD b 000
c 000 d 000 e 000 o

Fm-3m cubic, face-centered

0 12 --o

Ti-Co periodic edge

0 6 ooo

Ti-Ga intra-cell edge

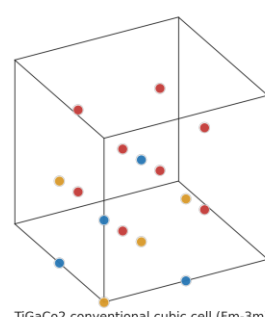
1 11 o--

Ti-Co periodic edge

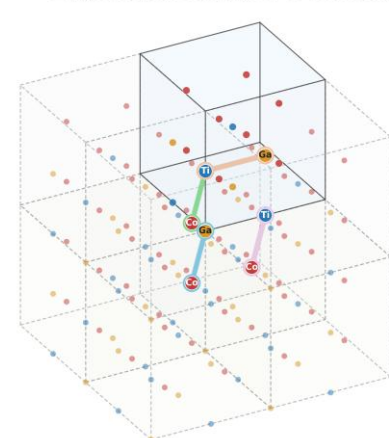
4 9 ---

Ga-Co periodic edge

Symmetry token → Fm-3m
(No. 225), cubic



Edge tokens mapped to periodic motifs



● Ti
● Ga
● Co
— Central cell
--- Image cells
0 12 --o Ti-Co
0 6 ooo Ti-Ga
1 11 o-- Ti-Co
4 9 --- Ga-Co